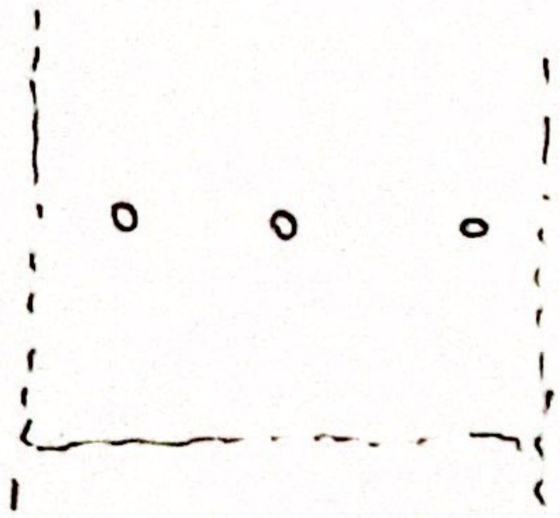
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150720823

Q1) $\lambda = 0.5\mu m$
$a = 0.1mm$
$d = 0.2mm$
$L = 5m$

$$y = \frac{m\lambda L}{a.d}$$



for $m=1$:

$$y_1 = \frac{1 \times 0.5\times10^{-6} \times 5}{0.1\times10^{-3} \times 0.2\times10^{-3}} = 12,5m$$

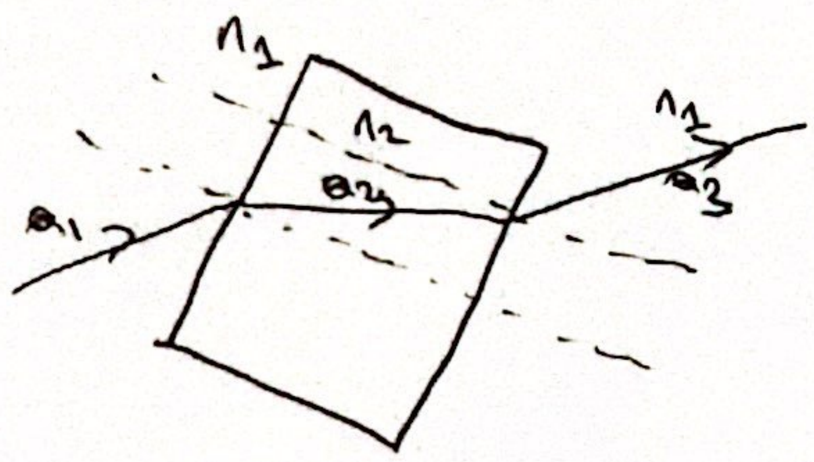
$$y_2 = \frac{2 \times 0.5\times10^{-6} \times 5}{0.1\times10^{-3} \times 0.2\times10^{-3}} = 25mm$$

$$y_3 = \frac{3 \times 0.5\times10^{-6} \times 5}{0.1\times10^{-3} \times 0.2\times10^{-3}} = 37,5mm$$

$$y_4 = \frac{4 \times 0.5\times10^{-6} \times 5}{0.1\times10^{-3} \times 0.2\times10^{-3}} = 50mm$$

$$y_5 = \frac{5 \times 0.5\times10^{-6} \times 5}{0.1\times10^{-3} \times 0.2\times10^{-3}} = 62,5m$$

Q3)



by using snells low

$$\frac{n_1}{n_2} = \frac{\sin\theta_2}{\sin\theta_1}$$

$$\frac{n_1}{n_2} = \frac{\sin\theta_2}{\sin\left(\sin^{-1}\left(\frac{n_2}{\sqrt{n_1^2+n_2^2}}\right)\right)}$$

$$\frac{n_1}{n_2} = \frac{\sin\theta_2}{n_2}\sqrt{n_1^2+n_2^2}$$

$$n_1 = \sin\theta_2\sqrt{n_1^2+n_2^2}$$

$$\sin\theta_2 = \frac{n_1}{\sqrt{n_1^2+n_2^2}}$$

$$\theta_2 = \sin^{-1}\left(\frac{n_1}{\sqrt{n_1^2+n_2^2}}\right)$$

$$\theta_2 = \tan^{-1}\left(\frac{n_1}{n_2}\right)$$

θ_2 is the Brewster's angle

* no reflection at the second interface.

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Q2) RED LED:
To ensure that the display is sufficiently bright, the red LED needs to have a high luminous intensity. The candela unit is used to measure this.

To chose the right power source for the red LED, one must take into account the forword voltage of the LED.

To achieve optimum performace and prevent overdriving or underdriving the LED, the operating current should be selected in accordance with the datasheet specifications of LED

GREEN LED: Like the red LED, the green LED needs to have a high luminous intensity to ensure that the display is sufficiently bright

similar to the red LED, the forword voltage drop and operating current of the green LED must be taken into account

Blue LED: The blue LED must likewise have a high luminous intensity. Think about the blue LED's operational current and forword voltage drop, to contribute to the desired color mix, the blue LED should have a restricted spectrical bandwidth centered on the blue wavelength

To make sure that the LEDs function within their suggested temperature range, the ambient temperature of the wall display area should be taken into acc. account. The employed LED drivers or power sources should be effective and dependable, giving the LED's a consistant and controlled voltage and current.

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Q4) Power = 1mW

diameter = 1mm

$U = Y_2 \varepsilon_0 E_0^2$

$S = UC$

$$S = \frac{1\times10^{-3}}{1\times\pi\times R^2} \qquad S = U\times C$$

$$\frac{1}{2}\varepsilon_0 E_0^2 \times C = \frac{1\times10^{-3}}{\pi(0.5\times10^{-3})^2}$$

$\varepsilon_0 = 8.85\times10^{-12}$

$$E_0^2 = \frac{2\times10^{-3}}{\pi\times0.5^2\times10^{-6}\times\varepsilon_0\times3\times10^8} = 0.095\,96\times10^7$$

$E_0 = 0.929\,59\ KN/C$

Q5) 1) Reduce capacitance: capacitance in the photodiode slows down the response time.

2) Decrease the function area: Reducing the function are of the photodiode reduces the transit time of the generated carriers, resulting in faster response

3) minimize the thickness of the intrinsic layer.

the intrinsic layer seperates the p-type and n-type regions in the photodiode.

4) improve the material quality: Enhancing the material quality, such as reducing defects and impurities, can enchance carrier mobility and reduce recombination, resulting in faster response time.

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